

Basic Peripheral Usage

The development board file structure is as follows. In the

`/userdata/magicbox/basic_function_demo/` directory, there are preset use cases for basic peripherals. Execute the program to control the corresponding peripherals.

```
root@ubuntu:/userdata/magicbox# tree -L 2
.
├── app                                #Application function files
│   ├── ros_ws                        # ros2 workspace, basically contains all
application functions
│   └── Web_RDK_Performance_Node      #Resource monitoring application
├── basic_function_demo                #Basic peripheral usage examples (python)
│   ├── button.py                    #Button usage script
│   ├── imu.py                       # IMU usage script
│   ├── servo.py                     #Servo control script
│   └── ws2812b.py                   #Light strip control script
├── config                            #Configuration files required by app
│   └── .....
├── dep                               #Third-party function packages required
by app
│   ├── llama.cpp                    # llama.cpp
│   ├── matcha-icefall-zh-baker      # TTS voice model dependency
│   └── sherpa-onnx                  # sherpa-onnx
├── launch                            #Files related to launch functions
│   ├── lamp                         #Light control scripts required for
startup
│   ├── start.py                     #Control script for starting and stopping
all application functions
│   └── start.sh                      #Auto-start execution script
├── log                               # log files
└── voice                             #Audio files
    ├── robot.mp3
    └── wakeup.mp3
```

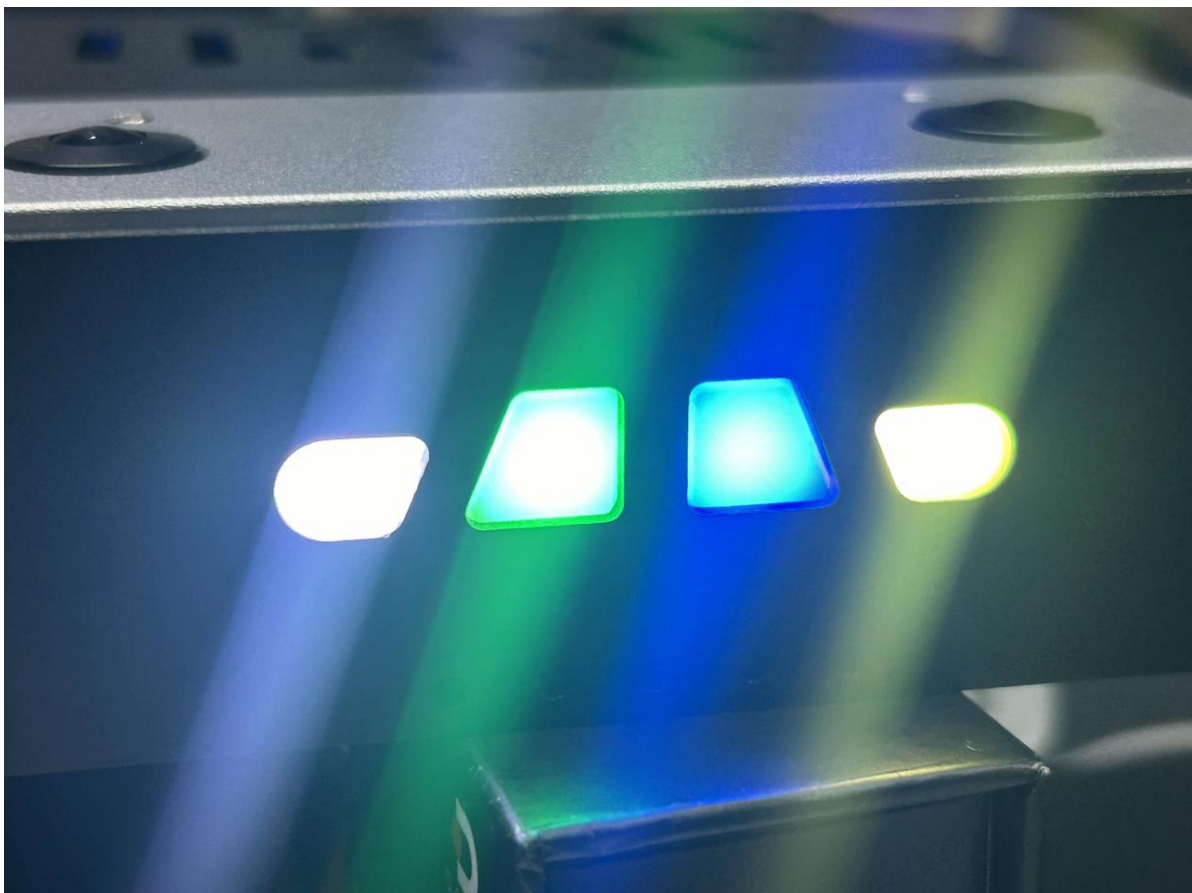
Servo Control

```
# Control the two support feet to extend downward and lift the device, based on
PWM implementation
python3 /userdata/magicbox/basic_function_demo/servo.py
```



Light Control

```
# Control the device light to turn on and display different colors, based on SPI  
implementation  
python3 python3 /userdata/magicbox/basic_function_demo/ws2812b.py
```



IMU Control

```
# Run IMU and print data, based on smbus2 implementation  
python3 /userdata/magicbox/basic_function_demo/imu.py
```

```
-----  
加速度 (m/s²): [-0.39743748 -0.02873042  9.72045874]  
角速度 (rad/s): [0.03342267  0.13582122  0.01731055]  
-----  
加速度 (m/s²): [-0.15801731  0.1484405  9.93114849]  
角速度 (rad/s): [-0.00452737  0.06138586 -0.0066579 ]  
-----  
加速度 (m/s²): [-0.30645781 -0.15322891  9.67257471]  
角速度 (rad/s): [0.13235911  0.02530003  0.00719053]
```

Button Control

```
# Run button press feedback, press the button and print button1 OK/button2  
OK/button3 OK, based on GPIO implementation  
python3 /userdata/magicbox/basic_function_demo/button.py
```